PROJECT / MIRPUR ROUTE LAB

Intelligent pathfinding on Mirpur, Dhaka road geometry

Version 3 | October 2026 | Education and algorithm comparison

01 PROBLEM AND DATA

Find paths between selected street-network coordinates while evaluating alternative graph-search methods.

OSM road geometry and directed road links come from an Overpass API query over Mirpur.

Traffic congestion and safety risk are deterministic SIMULATIONS; not observed conditions.

02 GRAPH AND MODEL

A directed graph G=(V,E) uses OSM road points as vertices and directed road segments as edges.

Base length d(u,v) is Haversine distance in kilometers. One-way tags inform edge direction.

Generalized cost c(u,v) = d(u,v)[1 + alpha*congestion(u,v) + beta*risk(u,v)].

The geographic heuristic h(n) is straight-line distance to the destination. Since c>=d,

h(n) is admissible and consistent under this nonnegative educational cost model.

03 IMPLEMENTED ALGORITHMS

BFS, DFS, depth-limited DFS, iterative deepening, UCS, directed bidirectional BFS,

Greedy Best-First, A*, Weighted A*, Beam Search. All support blocked-edge scenarios.

BFS, IDS and bidirectional BFS optimize edge count when they finish; UCS and A*

optimize selected nonnegative objective cost. Others need not be optimal.

04 INTERACTIVE EXPERIMENTS

Users select source/destination by clicking the map or choosing a location preset.

Controls tune simulated congestion, road-risk weight, road-block selection,

depth limit, beam width, and Weighted-A* coefficient.

Map replay marks expanded nodes, bounded frontier snapshots and current vertex.

Comparison reports distance, generalized cost, expansions, runtime and frontier size.

100/500-node deterministic induced-subgraph experiments may be disconnected.

05 LIMITATIONS AND VALIDATION

One-way streets are supported; turn restrictions, live travel speeds and safety evidence are not.

The server caches downloaded roads per warm process, not across all deployments.

Runtime is a single execution, not a statistically repeated benchmark.

Nine core tests passed locally; one Flask HTTP test was skipped because Flask was not installed.

Python compiled and client JavaScript parsed successfully. Online Overpass/Vercel tests remain.

06 REPRODUCIBILITY

Clone repository, install Flask>=3, run python -m unittest discover -s tests -v

and python api/index.py. Visit /api/health, /api/graph and the homepage.

For fair comparison keep endpoints and cost parameters fixed across algorithms.